



Q1. What are Zynq-7000 boards? What are the differences between Vivado Design Suite and PetaLinux associated with the Zynq-7000 board?

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A1. The Zynq-7000 family is based on System-on-Chip (SoC) architecture and incorporates a dual-core ARM Cortex-A9 with traditional Field Programmable Gate Arrays (FPGA) manufactured by Xilinx.

Vivado Design Suite is an automation tool for embedded systems, including Zynq-7000 boards. Vivado tool has a wide range of applications for designers, especially in embedded systems and RF domains. Vivado IDE is available for Windows and Linux operating systems. For more information, you can refer to



Fig. 1: Zynq-7000SoC board (Credit: xilinx.com)

the Vivado Design Suite user guide available on the Xilinx website.

PetaLinux offers all necessary requirements and tools to use embedded Linux solutions on Xilinx hardware systems. It supports development tools such as Zynq-7000 SoCs, MicroBlaze, and PowerPC boards. PetaLinux tools can run on any of the following Linux systems: Red Hat Enterprise Workstation/Server, CentOS, and Ubuntu.

For more details on PetaLinux, refer to PetaLinux Tools Documentation available on the Xilinx website.

Q2. How to interface inbuilt Bluetooth of ESP32 development board with Arduino IDE?

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A2. ESP32 board (refer Fig. 2) comes not only with Wi-Fi but also with classic Bluetooth and Bluetooth Low Energy (BLE). To interface the ESP32 development board with Arduino Uno, you can start with a simple application such as a temperature monitor using a 3-pin LM35 temperature sensor. The required software are Arduino IDE, Arduino code, and relevant libraries, including 'OneWire' for the temperature sensor and 'Arduino-esp32-master' for ESP32 board.

OneWire can be downloaded from Sketch/Include Library in Arduino IDE. Arduino-esp32-master can be downloaded from <https://github.com/espressif/arduino-esp32>.

Unzip Arduino-esp32-master.zip folder, copy and then paste BluetoothSerial folder under Arduino/Examples on your system. Open SerialToSerialBT from Arduino IDE, as shown in Fig. 3. Upload it to ESP32 board. This code establishes Bluetooth communication between

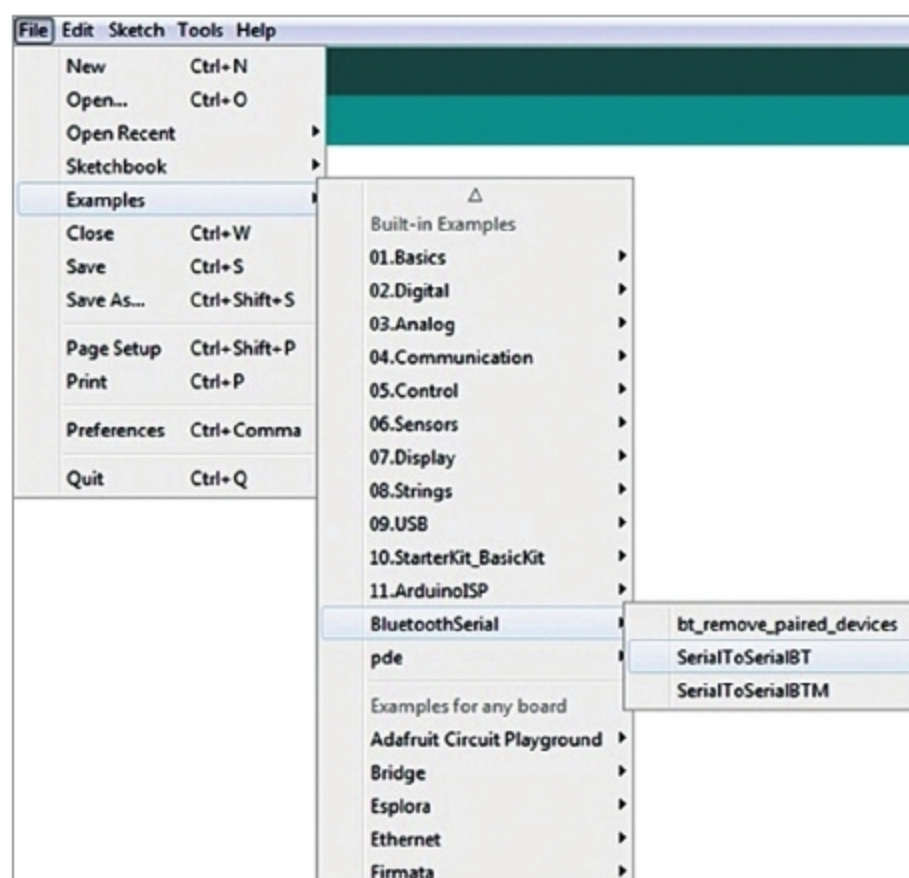


Fig. 3: Arduino IDE



Fig. 2: ESP32 board

ESP32 and your mobile phone.

Now, install a Bluetooth application on your Android phone, such as Serial Bluetooth Terminal available on the Play Store. You can also write a simple Arduino code for data exchange with temperature sensor. For more details, you may refer to the tutorial given at <https://www.instructables.com/id/How-to-Use-BluetoothBLE-With-ESP32/>.